Pressure pad power

Team Number:	203
Project Name:	**Pressure Pad Subsystem**
Team Member Names:	**Ramon Duarte**
Version:	**1.0**

All Major Components	Component Name	Part Number	SupplyVoltageRange
	Microcontroller	PIC18F57Q10	1.8 - 5.5 V
	Quad op-amp	MCP6004	1.8 - 6.0 V
	Indicator LED	D1 (generic LED)	5 V rail (330 ohm series R)
	Pressure pad / force sensor	FSR01CE	5 V circuit

N/A - no +12V rail	Component Name	Part Number	SupplyVoltageRange
c1. Regulator or Source Choice	N/A		

+5V Power Rail	Component Name	Part Number	SupplyVoltageRange
	Microcontroller	PIC18F57Q10	1.8 - 5.5 V
	Quad op-amp	MCP6004	1.8 - 6.0 V
	Indicator LED	D1 (generic LED)	5 V rail (330 ohm series R)
	Pressure pad / force sensor	FSR01CE	5 V circuit

c2. Regulator or Source Choice	+5V Regulator	MC7805CTG	7 - 35 V input
N/A - no -5V rail	**Component Name**	**Part Number**	**SupplyVoltageRange**
c - Regulator or Source Choice	N/A		
N/A - no +3.3V rail	**Component Name**	**Part Number**	**SupplyVoltageRange**
c - Regulator or Source Choice	N/A		
External Power Source 1	**Component Name**	**Part Number**	**SupplyVoltageRange**

External Power Source 1	Component Name	Part Number	SupplyVoltageRange
Power Source 1 Selection	AC/DC adapter to barrel jack	ALT-0601	100-240 VAC, 50/60 Hz
	+5V Regulator	MC7805CTG	7 - 35 V input
Estimated current required from source			
Total Remaining Current Available on External Power Source 1			
External Power Source 2	N/A - only one source required		
E. Calculate Battery Life (if applicable).	**N/A - subsystem uses external DC wall supply**		
Notes			

r budget

#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
1	250	250	mA
1	0.68	0.68	mA
1	20	20	mA
1	0.5	0.5	mA
		0	mA
		0	mA

#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
		0	
		0	
		0	
		0	
		0	
	Subtotal	0	mA
	Safety Margin	25%	
	Total Current Required on +12V Rail	0	mA
		0	mA
	Total Remaining Current Available on +12V Rail	0	mA

#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
1	250	250	mA
1	0.68	0.68	mA
1	20	20	mA
1	0.5	0.5	mA

		0	mA
	Subtotal	271.18	mA
	Safety Margin	25%	
	Total Current Required on +5V Rail	338.975	mA
1	1000	1000	mA
	Total Remaining Current Available on +5V Rail	661.025	mA
#	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
		0	
		0	
		0	
		0	mA
	Total Remaining Current Available on -5V Rail	0	mA
#	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**
		0	
		0	
		0	
		0	mA
	Total Remaining Current Available on +3.3V Rail	0	mA
Output Voltage	**AbsoluteMaximumCurrent (mA)**	**TotalCurrent(mA)**	**Unit**

Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
6 VDC	1000	1000	mA
		0	
1	1000	1000	mA
		0	
		342.175	**mA**
		657.825	**mA**